METHOD STATEMENT

# Acid Lock Pro — Integrated Chemical-Resistant Epoxy Lining System

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| **Project Name** | | **Document No.** | OMX-SMS-01 |
| **Main Contractor** | | **Revision** | 01 |
| **Consultant** | | **Date of Issue** | Nov 2026 |
| **System** | Acid Lock Pro | **Status** | For Approval |

## 1. Purpose

This method statement defines the sequence, resources, inspection and test plan and safety measures for the installation of the Acid Lock Pro system on site. It describes how each layer of the system is prepared, mixed, applied, inspected and protected so that the completed system achieves the performance stated in the O MIX Technical Data Sheets of its components.

## 2. Scope of Work

Acid Lock Pro is a heavy-duty, integrated epoxy lining system for concrete and steel exposed to aggressive chemicals. A surface-tolerant epoxy primer is followed by a solvent-free epoxy body coat and finished with a high-build novolac epoxy top coat that gives very high resistance to acids, alkalis, solvents and fuels.

- Substrate inspection, testing and acceptance.
- Surface preparation and repair of defects.
- Stage 1: Primer — O-Prime ST.
- Stage 2: Body coat — O-Guard EP.
- Stage 3: Chemical-resistant top coat (2 coats) — Mixcoat Novolac.
- Curing, protection, final testing and handover.

### 2.1 System build-up

| Stage | Product | Function | Consumption |
|---|---|---|---|
| **1** | O-Prime ST | Primer | 0.18 – 0.20 kg/m² |
| **2** | O-Guard EP | Body coat | 0.25 – 0.30 kg/m² |
| **3** | Mixcoat Novolac | Chemical-resistant top coat (2 coats) | 0.50 – 0.60 kg/m² (smooth) |

### 2.2 Typical areas of use

- Chemical plants and containment bunds.
- Wastewater and potable water tanks.
- Battery rooms and acid areas.
- Food, dairy and beverage plants.
- Heavy workshops and process areas.

## 3. Definitions

| Term | Meaning |
|---|---|
| **System** | All layers of the build-up applied in the stated sequence. |
| **Hold point** | Stage at which work stops until QA/QC (and the consultant, where required) approve. |
| **Recoat window** | Minimum and maximum time between two layers for proper inter-layer bonding. |
| **Dew point** | Temperature at which moisture condenses; the substrate must be at least 3 °C above it. |
| **Mock-up / trial area** | Area applied before the main works to approve materials, method and appearance. |
| **Pot life** | Time a mixed multi-component material remains workable. |
| **Holiday** | Pinhole or discontinuity in a lining, detected with a holiday detector. |
| **WFT / DFT** | Wet film thickness during application / dry film thickness after cure. |

## 4. Roles & Responsibilities

| Role | Responsibility |
|---|---|
| **Project Manager** | Planning, resources, approvals and coordination with the consultant and main contractor. |
| **Site Engineer** | Daily supervision; confirms substrate readiness, climate, mixing, consumption and recoat windows; keeps daily records. |
| **QA/QC Inspector** | Carries out tests at each hold point, records results and signs the inspection checklist. |
| **HSE Officer** | Toolbox talks, PPE, ventilation, safe access and storage; stops unsafe work. |
| **Applicator Foreman** | Leads the application team and follows this method statement; reports deviations. |
| **O MIX Technical Representative** | Technical support, site visits and advice on product use when requested. |

## 5. Materials

| Product | Description | Packaging |
|---|---|---|
| **O-Prime ST** | Surface-Tolerant Epoxy Primer for Steel, Light Metals & Concrete | 20 kg sets (Part A 16 kg + Part B 4 kg) and 5 kg sets (Part A 4 kg + Part B 1 kg) in metal pails. |
| **O-Guard EP** | Solvent-Free Polyamine Epoxy Lining for Potable Water Tanks | 5 kg sets: Part A 4.35 kg in pail + Part B 0.65 kg in jerrycan. |
| **Mixcoat Novolac** | Solvent-Free Novolac Epoxy Coating for Chemical-Resistant Floors and Linings | 5 kg and 20 kg sets: Part A in pail + Part B in jerrycan. |

### 5.1 Key product data

| Product | Mixing ratio | Pot life / working time | Recoat / drying |
|---|---|---|---|
| **O-Prime ST** | 4 : 1 by weight | 2 hours | Overcoat within the O-Prime ST recoat window (min. as per TDS) |
| **O-Guard EP** | 200 : 30 by weight — pre-weighed sets, mix as recommended | 40 min | 24 hours (max. 48 hours) |
| **Mixcoat Novolac** | 2 : 1 by volume — pre-weighed sets, mix full units as recommended | 30 – 40 min | Min. 16 hours / max. 48 hours |

## 6. Tools & Equipment

- Diamond grinder and/or shot blaster with dust extraction; industrial vacuum
- Slow-speed drill mixer (300 – 500 rpm) with spiral paddle
- Rollers, brushes, squeegees, notched trowels and steel trowels as required by each stage
- Scales and measuring containers for part mixing (where permitted)
- Thermo-hygrometer, surface thermometer and dew-point calculator
- Moisture meter and plastic sheets for moisture testing
- Masking tape, protective sheeting, barriers and warning signs
- Lighting and extension cables with residual current protection
- Wet-film comb gauge
- Pull-off adhesion tester
- Spiked roller and spiked shoes
- Holiday detector (wet sponge or high-voltage spark tester)
- Ultrasonic / magnetic DFT gauge
- Airless spray pump where specified
- Forced ventilation and gas detector for confined spaces

### 6.1 Personal protective equipment

- Safety shoes, helmet and high-visibility vest
- Chemical-resistant gloves (nitrile) and safety goggles
- Coveralls with long sleeves
- Dust masks (P2) during grinding and powder mixing
- Respirators with organic vapour cartridges for solvent-containing products and confined spaces
- Knee pads for floor work

## 7. Storage & Handling

- Store all products in original, unopened packaging in a dry, shaded and ventilated store at the temperatures stated in each TDS.
- Keep Part A, Part B (and Part C) of each kit together; do not mix components from different batches.
- Protect liquids from frost and direct sun, and powders from moisture.
- Store solvent-containing products away from heat, sparks and ignition sources.

- Use materials on a first-in, first-out basis and record batch numbers for each area.
- Condition materials to +15 °C to +25 °C for 24 hours before use in hot or cold weather.

## 8. Pre-Start Requirements

- Approved material submittal, this method statement and the inspection and test plan.
- Approved drawings showing areas, joints, upstands, drains and details.
- Work area handed over clean, free of other trades, with safe access, lighting and power.
- Mock-up / trial area executed and approved for system, colour and finish.
- Toolbox talk held with the team on this method statement and HSE measures.
- Climate monitoring equipment calibrated and available.

## 9. Substrate Inspection & Acceptance (Hold Point)

| Check | Acceptance criteria | Method |
|---|---|---|
| **Concrete age** | ≥ 28 days (cement-based substrates) | Site records |
| **Compressive strength** | ≥ 25 N/mm² | Rebound hammer (ASTM C805) or cores |
| **Surface tensile (pull-off) strength** | ≥ 1.5 N/mm² | Pull-off test ASTM D7234 |
| **Moisture content** | < 4 % (no moisture under plastic sheet after 16 h) | Moisture meter; plastic sheet test ASTM D4263 |
| **Steel surface cleanliness** | Sa 2½ (St 2 minimum where surface-tolerant primer is accepted) | ISO 8501-1 |
| **Dust on steel** | Rating ≤ 2 | ISO 8502-3 tape test |
| **Climate** | Substrate ≥ 3 °C above dew point; RH ≤ 85 % | ISO 8502-4 / thermo-hygrometer |

## 10. Surface Preparation

1. Remove oil and grease by solvent cleaning (SSPC-SP 1).
2. Abrasive blast-clean steel to Sa 2½ (or power-tool clean to St 2 where accepted for the surface-tolerant primer).
3. Remove dust; check dust rating and soluble salts where required.
4. Concrete: shot blast or diamond grind to remove laitance and open the pores.
5. Mask and protect adjacent surfaces, drains and fixtures.
6. Record substrate test results and obtain QA/QC approval before priming (hold point).

## 11. Application Sequence

Apply the system stage by stage in the order below. Each stage must be inspected and approved before the next stage starts, and the next stage must be applied within the recoat window stated.

### 11.1 Stage 1 — Primer: O-Prime ST

Consumption: 0.18 – 0.20 kg/m². Time before next stage: Overcoat within the O-Prime ST recoat window (min. as per TDS).

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Stir Part A thoroughly to disperse any settled pigment.
3. Add the complete Part B to Part A (4 : 1 by weight).
4. Mix 2 – 3 minutes with a slow-speed mixer until uniform in colour.
5. Add epoxy thinner only where needed for spray application, within the recommended limit.
6. Apply a stripe coat by brush on welds, edges, bolts and corners.
7. Apply the full coat by brush, roller or airless spray at 0.18 – 0.20 kg/m² (100 µm DFT) or to the DFT specified for the system.
8. Measure wet film thickness continuously and adjust application to reach the specified DFT.
9. Apply additional coats where the specified total DFT exceeds the single-coat thickness.

### 11.2 Stage 2 — Body coat: O-Guard EP

Consumption: 0.25 – 0.30 kg/m². Time before next stage: Min. 24 h, max. 48 h.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Add Part B to Part A (200 : 30 by weight) and mix 2 – 3 min with a slow-speed mixer.
3. Apply the body coat by roller or airless spray at 0.25 – 0.30 kg/m².
4. Apply the top coat after 24 hours and within 48 hours.
5. Check wet film thickness during application.

### 11.3 Stage 3 — Chemical-resistant top coat (2 coats): Mixcoat Novolac

Consumption: 0.50 – 0.60 kg/m² (smooth). Time before next stage: Min. 16 h, max. 48 h between coats; full chemical resistance after 7 days.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Empty the full Part B into the Part A container (2 : 1 by volume).
3. Mix 3 minutes at 300–400 rpm with a helical paddle, scraping sides and bottom.
4. Do not split kits unless weighed on calibrated scales; do not thin.
5. Apply the primer at 0.15 – 0.20 kg/m² and allow to cure.
6. Apply Mixcoat Novolac evenly by short-nap roller, brush or airless spray, working continuously.
7. For anti-slip finish, broadcast quartz sand into the first coat within 15 minutes; remove excess after cure.
8. Apply the second coat within the recoat window (16 – 48 hours); if exceeded, abrade first.

## 12. Hold Points & Recoat Windows

| Stage | Before starting | Waiting time before next stage | Approved by |
|---|---|---|---|
| **1. O-Prime ST** | Substrate accepted (Section 9) | Overcoat within the O-Prime ST recoat window (min. as per TDS) | QA/QC |

| Stage | Before starting | Waiting time before next stage | Approved by |
|---|---|---|---|
| **2. O-Guard EP** | Previous stage inspected and approved | Min. 24 h, max. 48 h | QA/QC |
| **3. Mixcoat Novolac** | Previous stage inspected and approved | Min. 16 h, max. 48 h between coats; full chemical resistance after 7 days | QA/QC |

## 13. Curing, Protection & Return to Service

- Protect each layer from rain, dew, dust, water and traffic until it has cured.
- Barricade the area and display warning signs during application and curing.
- Keep ventilation running in enclosed areas until full cure.
- Do not fill or immerse the lining before full cure (7 days at 25 °C).

## 14. Quality Control & Inspection Test Plan

| Activity | Inspection / Test | Frequency | Record |
|---|---|---|---|
| **Material receipt** | Product, batch, expiry, packaging condition | Each delivery | Material inspection report |
| **Climate** | Air / substrate temperature, RH, dew point (ISO 8502-4) | Start of shift and every 4 h | Daily climate log |
| **Substrate** | Strength, pull-off (ASTM D7234), moisture (ASTM D4263) | Each area (hold point) | Inspection request |
| **Mixing** | Complete kits, ratio, mixing time | Each mix | Daily record |
| **Film thickness** | WFT comb gauge (ASTM D4414) during application | Each coat | Daily record |
| **DFT** | Steel: SSPC-PA 2; concrete: ultrasonic gauge (SSPC-PA 9) | Each coat | DFT report |
| **Consumption** | Material used vs area | Each stage | Daily record |
| **Continuity** | Holiday test ASTM D4787 (concrete) / ASTM D5162 (steel) | 100 % of lining | Holiday test report |
| **Adhesion** | Pull-off test on trial area / as specified (ASTM D7234 or D4541) | As specified | Test report |
| **Final inspection** | Appearance, continuity, finish, details | Each area | Handover checklist |

## 15. Defects & Remedies

| Defect | Likely cause | Remedy |
|---|---|---|
| **Poor adhesion / delamination** | Weak, dusty or damp substrate; recoat window exceeded | Remove, re-prepare and reapply; abrade and re-prime if window exceeded |
| **Pinholes / bubbles** | Outgassing; air trapped in mixing; thick application | Seal porous areas; mix slowly; spiked roller; sand and recoat |
| **Blushing / tacky surface** | High humidity or low temperature during cure | Wash, abrade and recoat in suitable conditions |

| Defect | Likely cause | Remedy |
|---|---|---|
| **Soft spots** | Incorrect ratio or incomplete mixing | Remove and reapply with complete kits |
| **Holidays found** | Insufficient thickness; pinholes | Abrade locally, recoat and re-test |
| **Colour / shade variation** | Different batches; uneven application | Use one batch per area; apply additional coat |

## 16. Health, Safety & Environment

| Hazard | Control measures |
|---|---|
| **Skin / eye contact with resins and cement** | Gloves, goggles, coveralls; eye-wash station |
| **Inhalation of dust and vapours** | Dust extraction; ventilation; masks / respirators |
| **Fire (solvent vapours)** | No smoking or hot work; fire extinguisher at the work area |
| **Slips and trips** | Housekeeping; barricades and warning signs |
| **Manual handling** | Correct lifting; mechanical aids for drums and pails |
| **Electrical equipment** | Inspected tools; protected cables with RCD |
| **Confined spaces (tanks)** | Permit to work, gas testing, forced ventilation, standby person |

- Read the Safety Data Sheet of each product before use.
- Keep first-aid kit and eye-wash on site; follow the first-aid measures in each TDS.

### 16.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Allow small quantities of mixed resin to cure before disposal; dispose of according to local regulations.
- Do not discharge materials or washings into drains or soil.

## 17. Cleaning

- Clean tools of water-based and cementitious products with water immediately after use.
- Clean tools of epoxy and PU products with the recommended thinner before the material cures.
- Cured material can only be removed mechanically.

## 18. Limitations

- Do not apply outside the temperature and humidity limits stated in the TDS of each product.
- Do not apply on wet surfaces or when rain or dew is expected before the layer has cured.
- Do not mix or substitute products from other systems without written approval from O MIX.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

## Appendix A — Inspection Checklist

| Inspection Item | Result | Remarks |
|---|---|---|
| Approved submittal and method statement on site | ☐ Yes ☐ No | |
| Mock-up approved | ☐ Yes ☐ No | |
| Substrate tests passed (strength, pull-off, moisture) | ☐ Yes ☐ No | |
| Surface preparation approved | ☐ Yes ☐ No | |
| Stage 1 — O-Prime ST applied and approved | ☐ Yes ☐ No | |
| Stage 2 — O-Guard EP applied and approved | ☐ Yes ☐ No | |
| Stage 3 — Mixcoat Novolac applied and approved | ☐ Yes ☐ No | |
| Recoat windows respected | ☐ Yes ☐ No | |
| Final tests passed | ☐ Yes ☐ No | |
| Area protected and handed over | ☐ Yes ☐ No | |

| Inspected by | Signature | Date |
|---|---|---|
| | | |

## Appendix B — Daily Application Record

| Date | Area / zone | Stage | Batch No. | Qty used | Air / substrate °C | RH % | Sign |
|---|---|---|---|---|---|---|---|
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